

Lecture 8

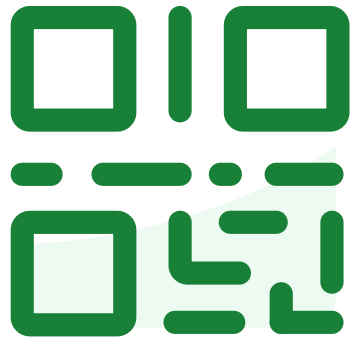
Linear Regression (3)

Probabilistic view of linear regression

EECS 189/289, Fall 2025 @ UC Berkeley

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Do not edit
How to change the
design



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Roadmap

- Recap of Regularization
- When Normal Equation Gets Tricky
- Least-Squares $\equiv$ Maximum Likelihood
- ~~Uncertainty $\approx$ Confidence Intervals~~
- ~~Decision Theory~~

Didn't get to cover in the lecture, not in scope for this lecture



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Recap of Regularization

- Recap of Regularization
- When Normal Equation Gets Tricky
- Least-Squares $\equiv$ Maximum Likelihood



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L2 Regularization (Ridge)

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \underbrace{\mathbf{E}_{\mathbf{D}}[\mathbf{w}]} + \underbrace{\lambda \mathbf{E}_{\mathbf{w}}[\mathbf{w}]}$$

$$\frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w}))^2$$

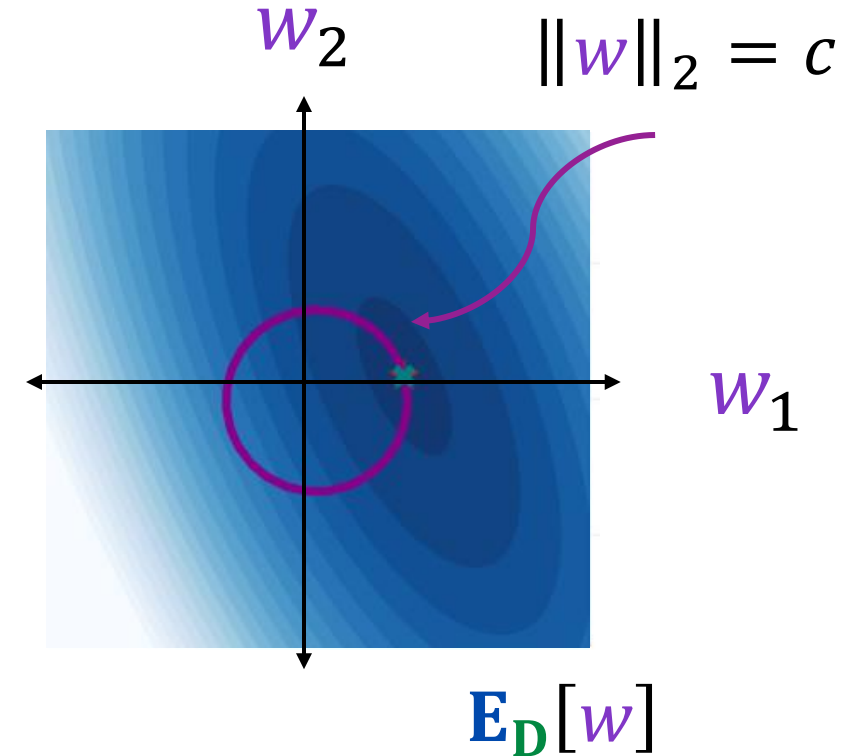
$$\frac{1}{2} L_2[\mathbf{w}]^2 = \frac{1}{2} \sum_d \mathbf{w}_d^2$$

$$\frac{1}{2} (\|\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w})\|_2)^2$$

$$\frac{1}{2} (\|e_n\|_2)^2$$

length of the
residual vector

$$L_2[\mathbf{w}] = \sqrt{\sum_d \mathbf{w}_d^2}$$



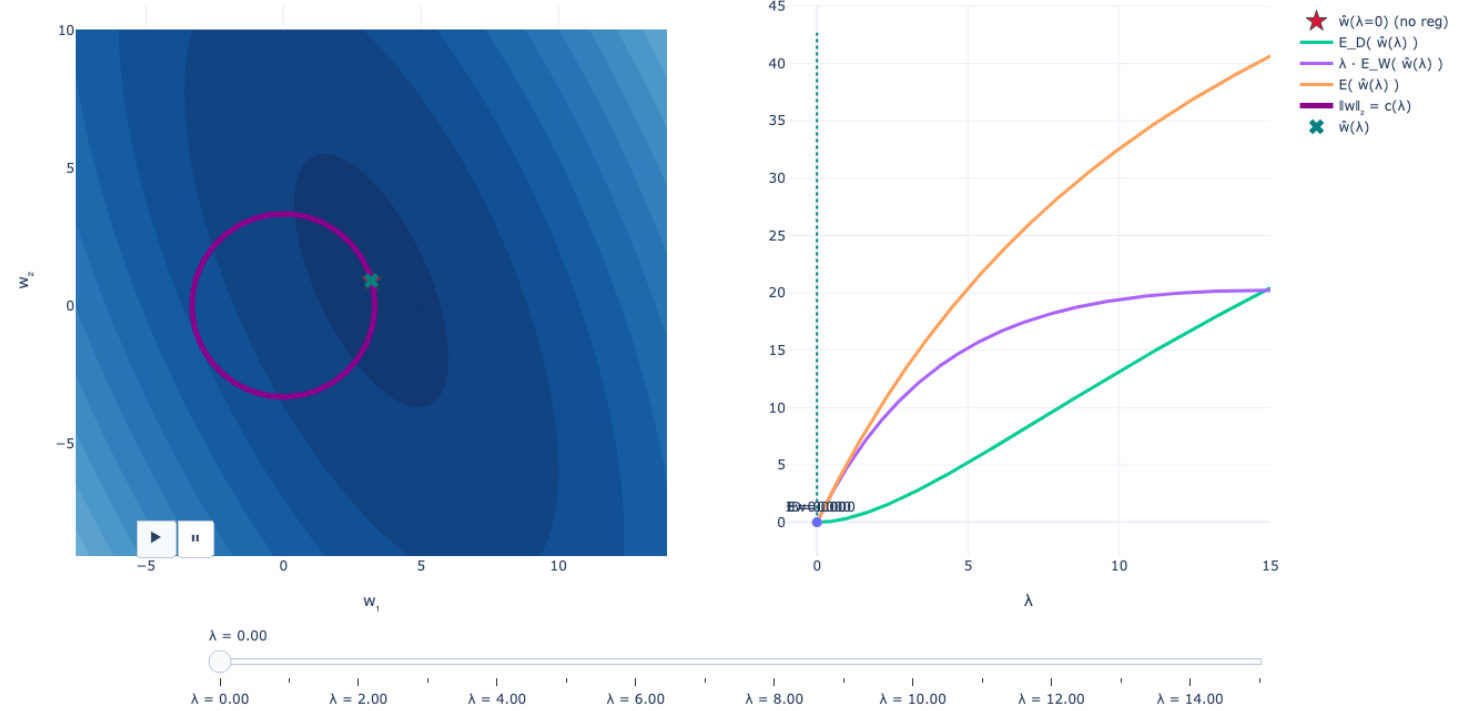
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L2 Regularization Demo

[Interactive link](#)



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L1 Regularization (Lasso)

$$L_1[\mathbf{w}] = \sum_j |\mathbf{w}_j|$$



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$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \underbrace{\mathbf{E}_{\mathbf{D}}[\mathbf{w}]} + \underbrace{\lambda \mathbf{E}_{\mathbf{w}}[\mathbf{w}]}$$

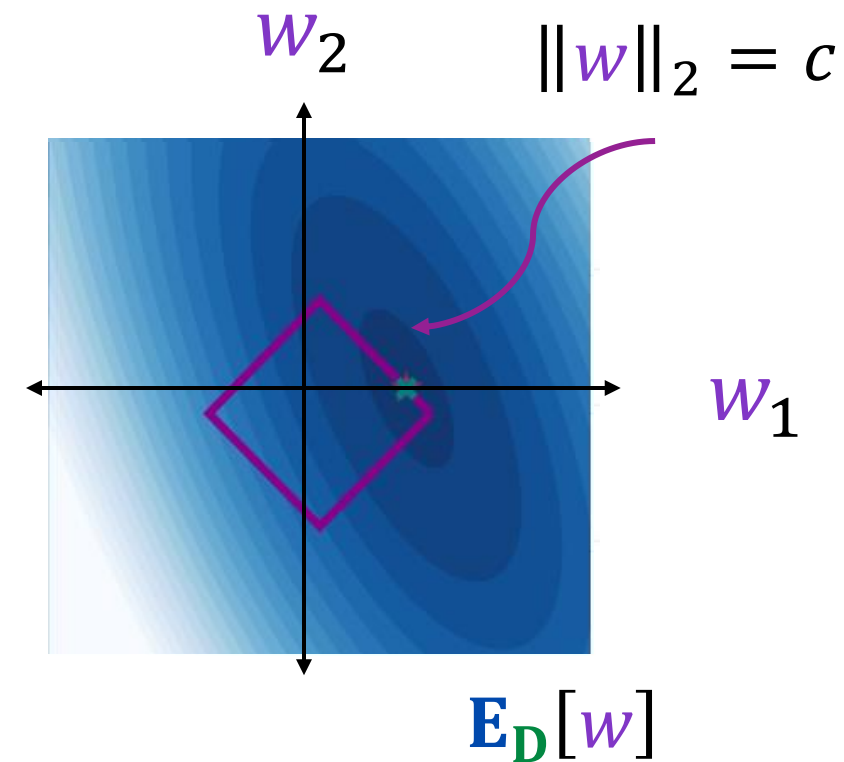
$$\frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w}))^2$$

$$\frac{1}{2} L_1[\mathbf{w}] = \frac{1}{2} \sum_d |\mathbf{w}_d|$$

$$\frac{1}{2} (\|\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w})\|_2)^2$$

$$\frac{1}{2} (\|e_n\|_2)^2$$

length of the
residual vector

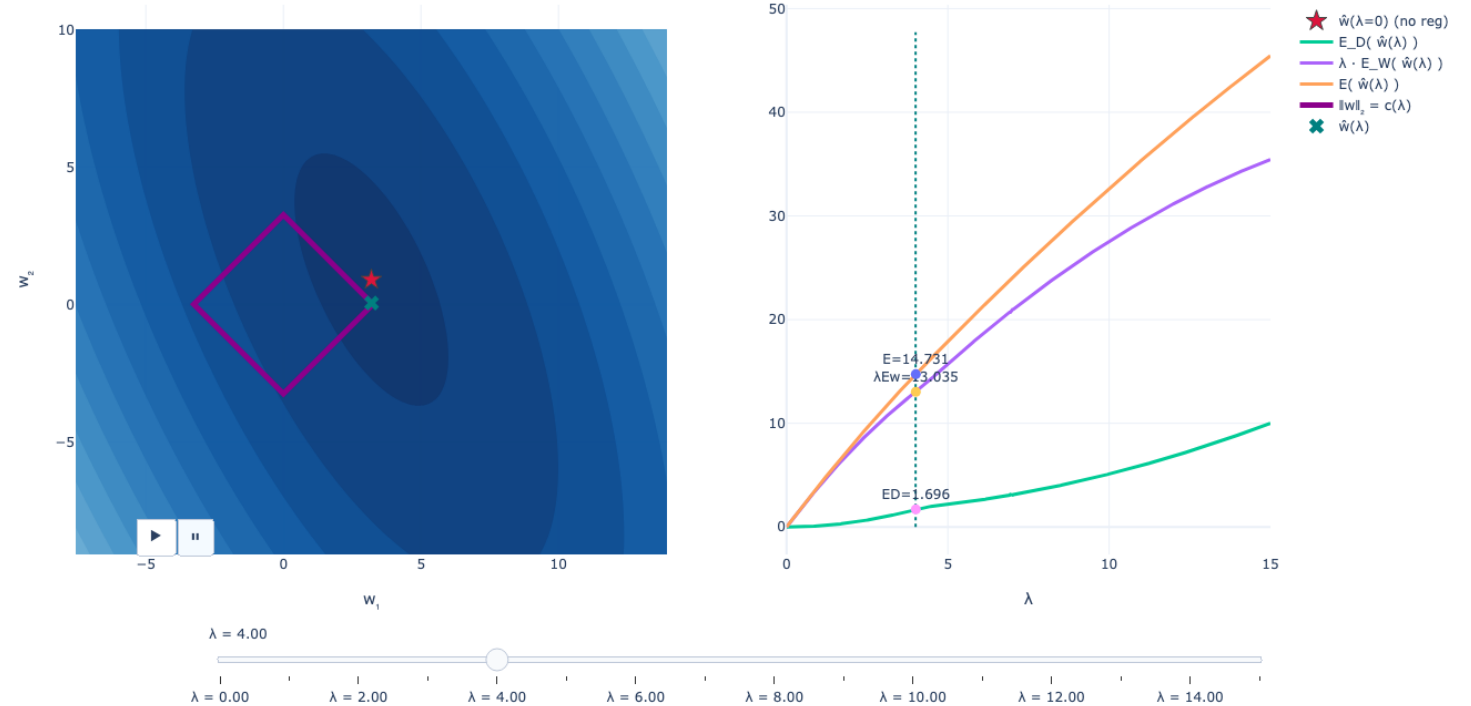


L1 Regularization Demo

[Interactive link](#)



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When Normal Equation Gets Tricky

- Recap of Regularization
- When Normal Equation Gets Tricky
- Least-Squares $\equiv$ Maximum Likelihood



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Normal Equation (Reminder)

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w}))^2 = \frac{1}{2} (\|\vec{e}\|_2)^2$$

$E(\mathbf{w})$ is minimum when $\vec{e}$ is orthogonal projection of $\mathbf{t}$.

Vector $\vec{e}$ is orthogonal to the $\text{span}(\mathbb{X})$, meaning that:

$$\mathbb{X}^T \vec{e} = 0$$

Adding the definition of residual

$$\mathbb{X}^T (\mathbf{t} - \mathbb{Y}(\mathbb{X}, \mathbf{w}^*)) = 0$$

Adding formula for $\mathbb{Y}(\mathbb{X}, \mathbf{w}^*) = \mathbb{X} \mathbf{w}^*$

$$\mathbb{X}^T \mathbf{t} - \mathbb{X}^T \mathbb{X} \mathbf{w}^* = 0$$

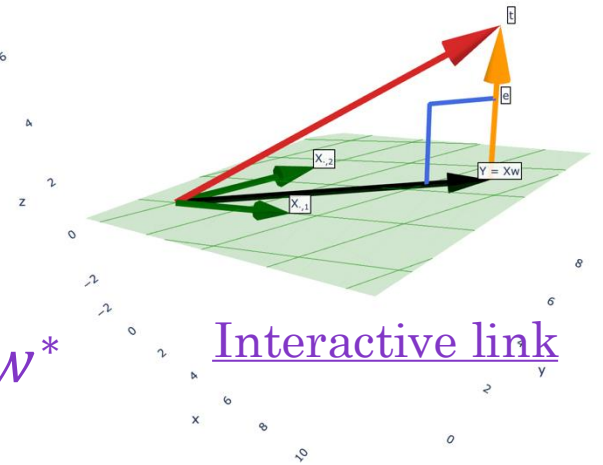
Moving terms

$$\mathbb{X}^T \mathbf{t} = \mathbb{X}^T \mathbb{X} \mathbf{w}^*$$

Normal
Equation

If $\mathbb{X}^T \mathbb{X}$ is
invertible

$$\mathbf{w}^* = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbf{t}$$



[Interactive link](#)



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When Normal Equation Gets Tricky

- When is $\mathbb{X}^T \mathbb{X}$ invertible?
 - Full column rank: $\text{rank}(\mathbb{X}) = D + 1$
 - Requires $N \geq D + 1$ (at least as many samples as features)
 - No perfect collinearity: $\det(\mathbb{X}^T \mathbb{X}) \neq 0$
 - All singular values must be non-zero: $\sigma_1, \sigma_2, \dots, \sigma_{D+1} > 0$.

Normal Equation

$$\mathbf{w}^* = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbf{t}$$
$$\mathbb{X} = \left[\begin{array}{c|ccc} & \overbrace{1 \quad x_{11} \quad \dots \quad x_{1D}}^{D+1} \\ 1 & \dots & \dots & \dots \\ 1 & x_{N1} & \dots & x_{ND} \end{array} \right] \left. \vphantom{\begin{array}{c} 1 \\ 1 \\ 1 \end{array}} \right\} N$$



If $N = D+1$ and X is full rank,
what is the training error?

Solution



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- If $N = D + 1$ and $\mathbb{X}$ is full rank, what is the training error?

$$\mathbb{X} \in \mathbb{R}^{N \times D+1}$$

$$\mathbb{X} \in \mathbb{R}^{D+1 \times D+1} \quad \text{Square matrix that is also full rank}$$

We can solve directly $\mathbb{X}^T \vec{e} = 0$

Multiply both sides by $(\mathbb{X}^T)^{-1}$ $\vec{e} = 0$

Training error will always be 0.



In which of the following situations is $X^T X$ likely to be singular or nearly singular?



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When Normal Equation Gets Tricky

- When is $\mathbb{X}^T \mathbb{X}$ invertible?

- Full column rank: $\text{rank}(\mathbb{X}) = D + 1$
- Requires $N \geq D + 1$ (at least as many samples as features)
- No perfect collinearity: $\det(\mathbb{X}^T \mathbb{X}) \neq 0$
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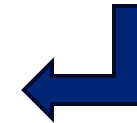
Normal Equation

$$\mathbf{w}^* = (\underbrace{\mathbb{X}^T \mathbb{X}}_{D+1})^{-1} \mathbb{X}^T \mathbf{t}$$
$$\mathbb{X} = \left[\begin{array}{cccc} 1 & x_{11} & \cdots & x_{1D} \\ 1 & \cdots & \cdots & \cdots \\ 1 & x_{N1} & \cdots & x_{ND} \end{array} \right] \left. \vphantom{\begin{array}{cccc} 1 & x_{11} & \cdots & x_{1D} \\ 1 & \cdots & \cdots & \cdots \\ 1 & x_{N1} & \cdots & x_{ND} \end{array}} \right\} N$$

- Ill-conditioning (numerical instability): Even if $\det(\mathbb{X}^T \mathbb{X}) \neq 0$

For any matrix A , $\det(A) = \prod_{i=1}^K \sigma_i$ where σ_i are eigenvalues.

When $\det(A) \approx 0$, means at least one singular value $\sigma_i \approx 0$.





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When Normal Equation Gets Tricky

Normal Equation

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{t}$$
$$\mathbf{X} = \left[\begin{array}{cccc} 1 & x_{11} & \cdots & x_{1D} \\ 1 & \cdots & \cdots & \cdots \\ 1 & x_{N1} & \cdots & x_{ND} \end{array} \right] \quad \left. \vphantom{\begin{array}{c} \mathbf{X} \\ \mathbf{X}^T \mathbf{X} \end{array}} \right\} \begin{array}{l} D+1 \\ N \end{array}$$

- When is $\mathbf{X}^T \mathbf{X}$ invertible?
 - Full column rank: $\text{rank}(\mathbf{X}) = D + 1$
 - Requires $N \geq D + 1$ (at least as many samples as features)
 - No perfect collinearity: $\det(\mathbf{X}^T \mathbf{X}) \neq 0$
 - All singular values must be non-zero: $\sigma_1, \sigma_2, \dots, \sigma_{D+1} > 0$.
- Ill-conditioning (numerical instability): Even if $\det(\mathbf{X}^T \mathbf{X}) \neq 0$
 - **Condition number** $\kappa = \frac{\sigma_{\max}}{\sigma_{\min}} \gg 10^8$ small noise in the data creates huge weight swing
 - Why?
 - Rounding error will be amplified when calculating $\kappa \varepsilon_{\text{machine}}$.



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Fix #1: Ridge Trick

- In Ridge regression, the closed form solution is

$$w^* = (X^T X + \lambda I)^{-1} X^T t$$

Proof in
discussion 4
this week

- Ridge replaces $X^T X$ in normal equation with $X^T X + \lambda I$.



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Fix #1: Ridge Trick

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Proof in
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- Ridge replaces $X^T X$ in normal equation with $X^T X + \lambda I$.

What are the eigenvalues of $X^T X + \lambda I$?

- A. σ_i
- B. $\sigma_i + \lambda$
- C. σ_i^2
- D. $\sigma_i^2 + \lambda$



**Which of the following options
are correct?**



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Fix #1: Ridge Trick

- In Ridge regression, the closed form solution is

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda I)^{-1} \mathbf{X}^T \mathbf{t}$$

Proof in
discussion 4
this week

- Ridge replaces $\mathbf{X}^T \mathbf{X}$ in normal equation with $\mathbf{X}^T \mathbf{X} + \lambda I$.
- This adds λ to every eigenvalue of $\mathbf{X}^T \mathbf{X}$:

$$\sigma_i^2 \rightarrow \sigma_i^2 + \lambda$$

- $\lambda > 0$ boosts all eigenvalues.
- Condition number shrinks:

$$\kappa_{\text{ridge}} = \frac{\sigma_{\max}^2 + \lambda}{\sigma_{\min}^2 + \lambda} \ll \kappa$$

This stabilizes
inversion

Illconditioning Example

With the condition factor that is large, due to very small eigenvalues resulting from very similar features, the weights become unstable.

Try changing the noise and you will see substantial swings in weights.

```
N, D = 500, 2 # one bias + 2 features
X = np.ones((N, D + 1)) # first column = bias
x1 = np.random.randn(N)
x2 = x1 + 1e-4 * np.random.randn(N) # almost identical feature
X[:, 1] = x1
X[:, 2] = x2
```

Condition $(X^T X) =$
420140223.27925146

```
t = 3 + 2*x1 - 1*x2 + 0.1*np.random.randn(N) # targets
```

```
XtX = X.T @ X
Xty = X.T @ t
w_ne = np.linalg.solve(XtX, Xty)
```

```
lam = 1e-2
w_ridge = np.linalg.solve(XtX + lam*np.eye(D+1), Xty)
```

```
w_neq: [ 3.00273423 12.87110771 -11.87562795]
```

```
w_ridge : [3.00275573 0.50070697 0.49481053]
```



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How to resolve the issue of ill-conditioned design matrix?



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SVD Recap: Stretch + Rotate

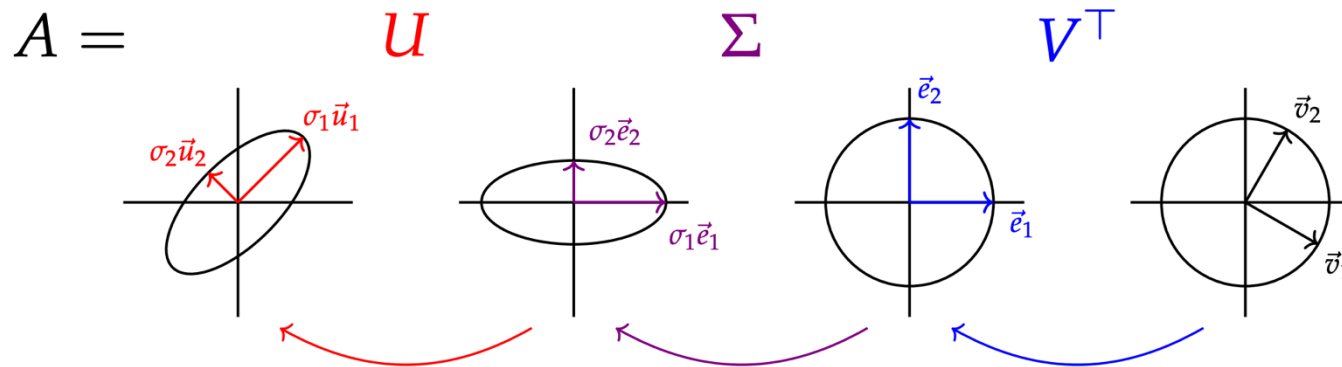
- Every real matrix $\mathbb{X}$ can be factored as

$$\mathbb{X} = U\Sigma V^T$$

U and V are rotations and have orthogonal columns ($U^T U = V^T V = I$).

Σ is diagonal scaling by singular values σ_i .

$\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$ with $r = \text{rank}(\mathbb{X})$.



Fix #2: Flipping Singular Values with Moore–Penrose Pseudo-Inverse



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- Ordinary inverse of $\mathbb{X} = U\Sigma V^T$ is $\mathbb{X}^{-1} = V\Sigma^{-1}U^T$

- If some $\sigma_i = 0$, we cannot invert directly.

- Moore–Penrose Pseudo-inverse:

$$\sigma_{ii}^+ = \begin{cases} \frac{1}{\sigma_i} & \text{if } \sigma_i > \varepsilon \sigma_{max} \\ 0 & \text{otherwise} \end{cases}$$

- We safely “flip” only the non-tiny singular values.

- Pseudo-Inverse:

$$\mathbb{X}^+ = V\Sigma^+U^T$$

Fix #2: Flipping Singular Values with Moore–Penrose Pseudo-Inverse



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- Pseudo-Inverse:

$$\mathbb{X}^+ = V\Sigma^+U^T$$

- Properties:

- $\mathbb{X}\mathbb{X}^+\mathbb{X} = \mathbb{X}$,
- $\mathbb{X}^+\mathbb{X}\mathbb{X}^+ = \mathbb{X}^+$,
- $(\mathbb{X}\mathbb{X}^+)^T = \mathbb{X}\mathbb{X}^+$,
- $(\mathbb{X}^+\mathbb{X})^T = \mathbb{X}^+\mathbb{X}$.

- These guarantee uniqueness of $\mathbb{X}^+$.



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$\mathbf{X}\mathbf{X}^+$ Is the Orthogonal Projector

• Let $P = \mathbf{X}\mathbf{X}^+$. Then:

1. **Projector:** $P^2 = \mathbf{X}\mathbf{X}^+\mathbf{X}\mathbf{X}^+ = \mathbf{X}\mathbf{X}^+ = P$

2. **Symmetric:** $P^T = P$

3. **Range:** $Py = \mathbf{X}\mathbf{X}^+\mathbf{X}w = \mathbf{X}w = y$

$\left\{ \begin{array}{l} P \text{ acts as identity on } \text{span}(\mathbf{X}). \\ \text{range}(P) \text{ is } \text{span}(\mathbf{X}). \end{array} \right.$

$P = \mathbf{X}\mathbf{X}^+$ is the orthogonal projector onto $\text{span}(\mathbf{X})$.

Obtaining w^*



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$P = \mathbb{X}\mathbb{X}^+$ is the orthogonal projector onto $\text{span}(\mathbb{X})$.



We will use P to find the projection of $\mathbf{t}$ onto the $\text{span}(\mathbb{X})$.

$$y = P\mathbf{t}$$

Replacing $P = \mathbb{X}\mathbb{X}^+$

$$y = \mathbb{X}\mathbb{X}^+\mathbf{t}$$

Knowing that $y = \mathbb{X}w$

$$w^* = \mathbb{X}^+\mathbf{t}$$

Very important result!!!

This derivation
is incorrect, see
next slide



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Obtaining w^*

$P = \mathbb{X}\mathbb{X}^+$ is the orthogonal projector onto $\text{span}(\mathbb{X})$.



We will use P to find the projection of $\mathbf{t}$ onto the $\text{span}(\mathbb{X})$.

$$y = P\mathbf{t} \quad \text{Replacing } P = \mathbb{X}\mathbb{X}^+$$

$$y = \mathbb{X}\mathbb{X}^+\mathbf{t} \quad \text{Knowing that } y = \mathbb{X}\mathbf{w}$$

$$\mathbb{X}\mathbf{w} = \mathbb{X}\mathbb{X}^+\mathbf{t} \longrightarrow \mathbf{w} = \mathbb{X}^+\mathbf{t} + (\mathbf{I} - \mathbb{X}\mathbb{X}^+)z \text{ for } z \in R^{D+1}$$

Using
minimum-norm
argument

$$\mathbf{w}^* = \mathbb{X}^+\mathbf{t} \quad \text{Very important result!!!}$$

$$\mathbf{w}^* = \arg \min_{\mathbf{w}: \mathbb{X}\mathbf{w} = y} \|\mathbf{w}\|_2^2$$



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Fix #2: Moore–Penrose Pseudo-Inverse

How can Penrose Pseudo-Inverse help us with solving normal equation?

- Moore–Penrose Pseudo-inverse:

$$\sigma_{ii}^+ = \begin{cases} \frac{1}{\sigma_i} & \text{if } \sigma_i > \varepsilon \sigma_{max} \\ 0 & \text{otherwise} \end{cases}$$

- We safely “flip” only the non-tiny singular values.

Closed-form solution for least squares:

$$\mathbf{w}^* = \mathbf{X}^+ \mathbf{t}$$

This works no matter whether $\mathbf{X}$ is tall ($N > D$), square, or wide ($N < D$). Also regardless of singular values.



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Obtaining w^*

$P = \mathbb{X}\mathbb{X}^+$ is the orthogonal projector onto $\text{span}(\mathbb{X})$.



We will use P to find the projection of t onto the $\text{span}(\mathbb{X})$.

$$y = Pt \quad \text{Replacing } P = \mathbb{X}\mathbb{X}^+$$

$$y = \mathbb{X}\mathbb{X}^+t \quad \text{Knowing that } y = \mathbb{X}w$$

$$\mathbb{X}w = \mathbb{X}\mathbb{X}^+t \longrightarrow w = \mathbb{X}^+t + (I - \mathbb{X}\mathbb{X}^+)z \text{ for } z \in R^{D+1}$$

Using
minimum-norm
argument

$$w^* = \mathbb{X}^+t \quad \text{Very important result!!!}$$

$$w^* = \arg \min_{w: \mathbb{X}w=y} \|w\|_2^2$$



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Why This w^* Amongst So Many?

If $\mathbb{X}$ is rank-deficient, there are infinitely many w with the same prediction y .

You can add any vector from the nullspace of $\mathbb{X}$.

The Pseudo-inverse picks the one with smallest Euclidean form:

$$w^* = \arg \min_{w: \mathbb{X}w=y} \|w\|_2^2$$



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Fix 2: Why It Helps?

Issue	How pseudo-inverse fixes it?
Rank deficiency (perfect multicollinearity)	Sets 0 for $\sigma = 0 \Rightarrow$ removes redundant directions, gives the <i>minimum-norm</i> weight vector.



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Fix 2: Why It Helps?

Issue	How pseudo-inverse fixes it?
Rank deficiency (perfect multicollinearity)	Sets 0 for $\sigma = 0 \Rightarrow$ removes redundant directions, gives the <i>minimum-norm</i> weight vector.
Ill-conditioning	Truncation threshold discards numerically meaningless directions, implicitly adds ridge-like regularization.



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Rank deficiency (perfect multicollinearity)	Sets 0 for $\sigma = 0 \Rightarrow$ removes redundant directions, gives the <i>minimum-norm</i> weight vector.
Ill-conditioning	Truncation threshold discards numerically meaningless directions, implicitly adds ridge-like regularization.
Underdetermined systems ($N < D$)	Returns the unique solution with the smallest L_2 -norm among infinitely many exact fits.



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Fix 2: Why It Helps?

Issue	How pseudo-inverse fixes it?
Rank deficiency (perfect multicollinearity)	Sets 0 for $\sigma = 0 \Rightarrow$ removes redundant directions, gives the <i>minimum-norm</i> weight vector.
Ill-conditioning	Truncation threshold discards numerically meaningless directions, implicitly adds ridge-like regularization.
Underdetermined systems ($N < D$)	Returns the unique solution with the smallest L_2 -norm among infinitely many exact fits.
Implementation convenience	<code>np.linalg.pinv</code> & <code>torch.pinverse</code> expose <code>rcond</code> so you control the σ -cut-off in one line of code.

Least-Squares $\equiv$ Maximum Likelihood

- Recap of Regularization
- When Normal Equation Gets Tricky
- Least-Squares $\equiv$ Maximum Likelihood
- Uncertainty $\approx$ Confidence Intervals
- Decision Theory



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Least Squares $\equiv$ Maximum Likelihood

Recall: We fit a linear featurized function by *Minimizing the sum-of-squares error*.

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (\mathbf{t}_n - y(\mathbf{x}_n, \mathbf{w}))^2$$

Is this just algebraic convenience ... or is there a probabilistic justification?



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Additive-Gaussian Noise Assumption

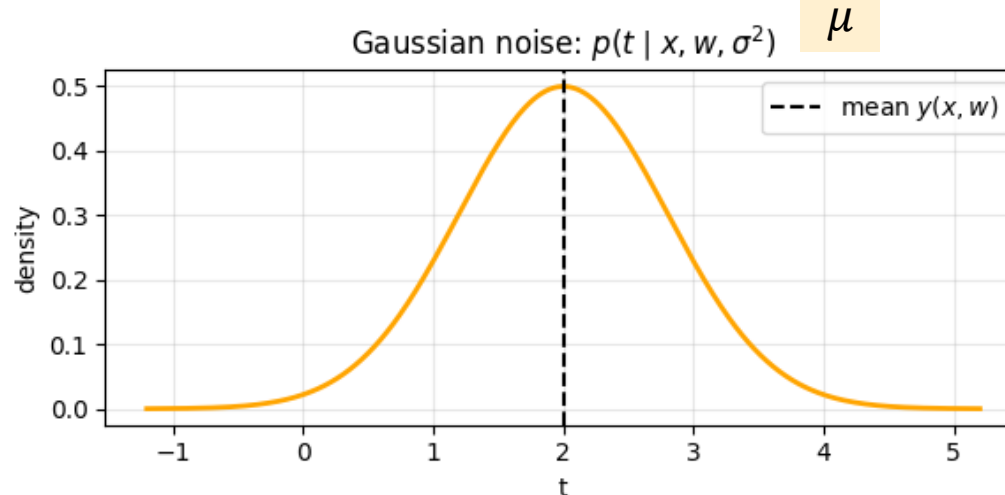
- Assume the target t is the true signal plus noise:

$$t = y(\mathbf{x}, \mathbf{w}) + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ (zero-mean Gaussian random variable) $\rightarrow$ i.i.d. across points.

- The conditional likelihood of one observation is:

$$p(t | \mathbf{x}, \mathbf{w}, \sigma^2) = \mathcal{N}(t | y(\mathbf{x}, \mathbf{w}), \sigma^2)$$





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Maximum Likelihood: Over the Dataset

- Consider the data set of i.i.d. inputs $\mathbb{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ with corresponding target values $t_1, \dots, t_N$. We group the target variables $\{t_n\}$ into a column vector that we denote by $\mathbf{t}$.

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{t}|y(\mathbb{X}, \mathbf{w}), \sigma^2)$$

Datapoints are
independent

$$= \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$



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Least Squares $\equiv$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

The optimum value of $\mathbf{w}$ in the above equation equals the optimum value of the log of the function (because log is a monotonically increasing function)

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = \sum_{n=1}^N \ln(\mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2))$$

Writing the equation for a Normal distribution

$$= \sum_{n=1}^N \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2}\right)$$

Applying $\log a \times b = \log a + \log b$

$$= \sum_{n=1}^N \left(\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{1}{2\sigma^2}(t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \right)$$

Simplify and separate two terms

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2$$



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Least Squares $\equiv$ Maximum Likelihood

$$\begin{aligned}\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) &= -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma^2) - \frac{1}{\sigma^2} E_D(\mathbf{w}) \quad (*)\end{aligned}$$

$$\rightarrow E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2$$

- The first two terms in can be treated as constants when determining $\mathbf{w}$ because they are independent of $\mathbf{w}$.
- Therefore, maximizing the likelihood function under a Gaussian noise distribution is equivalent to minimizing the sum-of-squares error function.

Maximum Likelihood



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$$\nabla_{\mathbf{w}} \ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{1}{\sigma^2} \nabla_{\mathbf{w}} E_{\mathbf{D}}(\mathbf{w})$$

Sum rule

$$= \nabla_{\mathbf{w}} \left(-\frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \right)$$

Chain rule

$$= -\frac{1}{2\sigma^2} \left(\sum_{n=1}^N \nabla_{\mathbf{w}} (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2 \right)$$

Simplify

$$= -\frac{1}{2\sigma^2} 2 \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n)) (-\phi(\mathbf{x}_n))$$

$$= \frac{1}{\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n)) \phi(\mathbf{x}_n)$$

Setting the gradient to 0 gives:

$$\frac{1}{\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n)) \phi(\mathbf{x}_n) = 0$$

$$\sum_{n=1}^N t_n \phi(\mathbf{x}_n) - \mathbf{w}^T \phi(\mathbf{x}_n) \phi(\mathbf{x}_n) = 0$$

$$\sum_{n=1}^N t_n \phi(\mathbf{x}_n) = \sum_{n=1}^N \boxed{\mathbf{w}^T \phi(\mathbf{x}_n)} \phi(\mathbf{x}_n) \quad \text{const}$$

$$\sum_{n=1}^N t_n \phi(\mathbf{x}_n) = \sum_{n=1}^N \phi(\mathbf{x}_n) \phi(\mathbf{x}_n)^T \mathbf{w}$$

Maximum Likelihood



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$$\sum_{n=1}^N t_n \phi(\mathbf{x}_n) = \sum_{n=1}^N \phi(\mathbf{x}_n) \phi(\mathbf{x}_n)^T \mathbf{w}$$

If we stack features $\phi(\mathbf{x}_n)$ row-wise in the design matrix $\Phi(\mathbb{X})$

$$\Phi(\mathbb{X}) = \begin{bmatrix} \phi(\mathbf{x}_1)^T \\ \phi(\mathbf{x}_2)^T \\ \dots \\ \phi(\mathbf{x}_N)^T \end{bmatrix}$$

$$\Phi(\mathbb{X})^T \mathbf{t} = \Phi(\mathbb{X})^T \Phi(\mathbb{X}) \mathbf{w}_{ML}$$

$$\mathbf{w}_{ML} = (\Phi(\mathbb{X})^T \Phi(\mathbb{X}))^{-1} \Phi(\mathbb{X})^T \mathbf{t}$$

Normal Equations for the least squared problem



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Least Squares $\equiv$ Maximum Likelihood

$$p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

$$\ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} E_D(\mathbf{w})$$

$$\text{with } E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (t_n - \mathbf{w}^T \phi(\mathbf{x}_n))^2$$

$$\nabla_{\mathbf{w}} \ln(p(\mathbf{t}|\mathbb{X}, \mathbf{w}, \sigma^2)) = -\frac{1}{\sigma^2} \nabla_{\mathbf{w}} E_D(\mathbf{w}) \longrightarrow \mathbf{w}_{ML} = (\Phi(\mathbb{X})^T \Phi(\mathbb{X}))^{-1} \Phi(\mathbb{X})^T \mathbf{t}$$

$$\Phi(\mathbb{X}) = \begin{bmatrix} \phi(\mathbf{x}_1)^T \\ \phi(\mathbf{x}_2)^T \\ \vdots \\ \phi(\mathbf{x}_N)^T \end{bmatrix}$$

Hence the least-squares solution is the **MLE** under Gaussian noise.

Lecture 8

Linear Regression (3)

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters: Chapter sections 4.1 and 4.2